

Siddharth Yadav

M.Tech in Computer Science and Engineering

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Education

Indian Institute of Technology, Patna	2025 – Present
M.Tech. (Computer Science and Engineering)	CGPA: 8.38 (Current)
Gurukula Kangri (Deemed to be University), Haridwar	2020 – 2024
B.Tech. (Computer Science and Engineering)	CGPA: 8.32
UP Board	2017
Secondary Education	89%

Achievements

- **GATE CSE 2025:** Secured AIR **1374** with a **99.20 percentile** among 1,70,825 candidates. [\[Scorecard\]](#)
- **Competitive Programming:** Solved **1000+** problems across [LeetCode](#) and [GeeksforGeeks](#).
- **Coding Profile:** Current [LeetCode](#) Rating: **1698**.
- **Advanced DSA:** Solved complex problems in Segment Trees, Tries (Prefix Tree), Graph Algorithms (MST), and DSU.

Positions of Responsibility

Teaching Assistant – Data Structures IIT Patna	Jan 2026 – Apr 2026
Teaching Assistant – C Programming IIT Patna	Aug 2025 – Nov 2025

Projects

Multi-Agent Travel Planner App using LangGraph and MCP	2026
<i>Python, LangGraph, Model Context Protocol (MCP), Groq, Tavily, AviationStack</i>	
• Designed a LangGraph-based multi-agent workflow for travel planning with state-driven orchestration and conditional tool execution. • Integrated Tavily, Weather, and AviationStack MCP servers using <code>MultiServerMCPClient</code> for real-time web search, weather forecasts, and flight information retrieval.	
Scalable URL Shortener with Redis Caching & Rate Limiting	2026
<i>Node.js, Express.js, MongoDB, Redis, Base62 Encoding</i>	
• Designed and implemented a scalable URL shortening service supporting high read traffic with efficient key-based lookups. • Built RESTful APIs to generate unique short URLs using Base62 encoding, ensuring collision-free mappings. • Improved redirection latency by integrating Redis caching, significantly reducing database query load. • Implemented IP-based and user-based rate limiting to prevent system abuse under high traffic spikes.	
DocChat: Multi-Agent RAG System	2025
<i>Python, LangGraph, ChromaDB, Docling, IBM watsonx AI, Gradio</i>	
• Designed a multi-agent RAG pipeline using LangGraph for document ingestion, retrieval, response generation, and answer verification. • Integrated ChromaDB vector search and hybrid retrieval to improve contextual retrieval accuracy for complex queries. • Built a Gradio conversational interface supporting document uploads and grounded AI responses powered by IBM watsonx AI.	
AI-Powered Job Application Tracker	2025
<i>React.js, Node.js, Express.js, MongoDB, Groq AI, JWT</i>	
• Built a full-stack Kanban board web application to track job applications across stages (Applied, Phone Screen, Interview, Offer, Rejected). • Integrated Groq AI to parse job descriptions and extract company, role, required skills, location, and seniority using prompt engineering. • Developed AI-generated resume bullet point suggestions tailored to each specific job description. • Implemented a secure JWT-based authentication system with encrypted login and registration flows.	
IPL Winner Prediction using Machine Learning	2024
<i>Python, Pandas, NumPy, scikit-learn, Matplotlib</i>	
• Developed a predictive machine learning model to estimate match winners using historical IPL datasets and performance statistics. • Evaluated Logistic Regression and Random Forest algorithms, achieving over 80% prediction accuracy.	

Technical Skills

Languages	C, C++, Python, SQL, HTML, CSS
AI / GenAI	Generative AI, LangChain, LangGraph, Multi-Agent Systems, RAG Architectures, Model Context Protocol (MCP), LLMs, SLMs, Prompt & Context Engineering, Model Fine-Tuning
Machine Learning	Pandas, NumPy, scikit-learn
Web & Backend	Node.js, Express.js, React.js, REST APIs, JWT
Databases	Redis, MySQL, MongoDB, ChromaDB, Vector Databases, HNSW
Core CS	Data Structures & Algorithms, Object-Oriented Programming (OOP), Multithreading, Multiprocessing, Thread Synchronization, SDLC, Spec-Driven Development (SDD)
Networking & Cloud	TCP/IP, HTTP/HTTPS, DNS, Data Transmission, AWS (EC2, S3, IAM basics)
Tools & OS	Git, GitHub, Linux, Windows

Certifications

- **Agentic AI Certified Foundations Associate** | *Oracle* [\[Credential\]](#)
- **Introduction to Model Context Protocol** | *Claude Academy* [\[Credential\]](#)
- **Model Context Protocol: Advanced topics** | *Claude Academy* [\[Credential\]](#)
- **Agentic AI with LangChain and LangGraph** | *IBM (Coursera)* [\[Credential\]](#)
- **Build AI Agents using MCP** | *IBM (Coursera)* [\[Credential\]](#)
- **Vector Databases for RAG: An Introduction** | *IBM (Coursera)* [\[Credential\]](#)
- **System Design - Skill Up** | *GeeksforGeeks* [\[Credential\]](#)
- **Generative AI, LLM & RAG - Skill Up** | *GeeksforGeeks* [\[Credential\]](#)
- **Python Skill Up** | *GeeksforGeeks* [\[Credential\]](#)
- **C++ Skill Up** | *GeeksforGeeks* [\[Credential\]](#)
- **C Skill Up** | *GeeksforGeeks* [\[Credential\]](#)
- **Git - Skill Up** | *GeeksforGeeks* [\[Credential\]](#)

Key Courses Taken

Data Structures, Algorithms, DBMS, Machine Learning, Artificial Intelligence, Theory of Computation, Operating Systems, Computer Architecture and Organization.